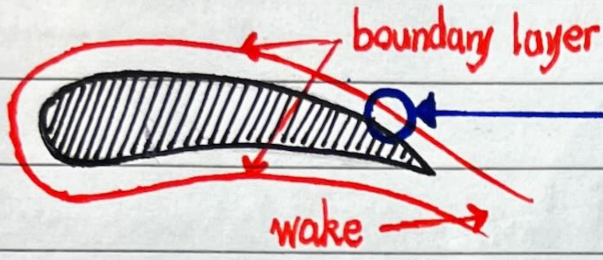
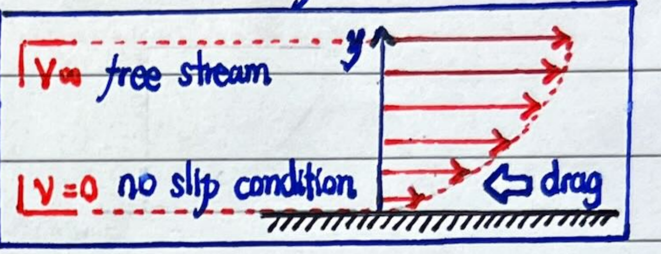
11/19/2022 Pipe Flow Pre-Lab

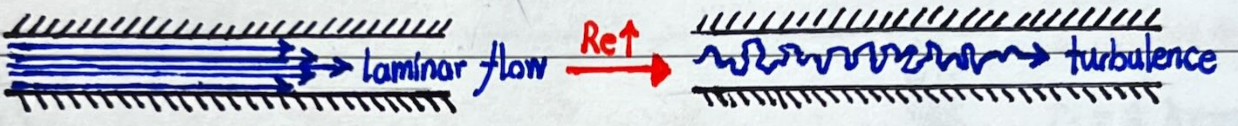
Introduction

- effect of soild walls on fluid flows
  - the interaction of flows with curved surfaces
  - effects of the boundary layer generated

better understand real world phenomena



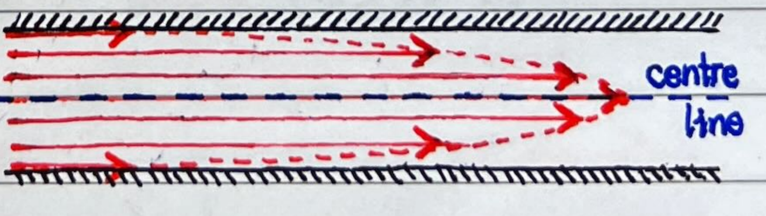




- examine the transition from laminar to turbulent
  - measuring the pressure drop along the length of the pipe
  - allow us to compute the shear stress, τ
- investigate the C_f skin friction coefficient varies with Re_D
- deduce the Re_{crit} critical Reynolds number
- compare results to theory and past findings

Theory

fully developed pipe flow



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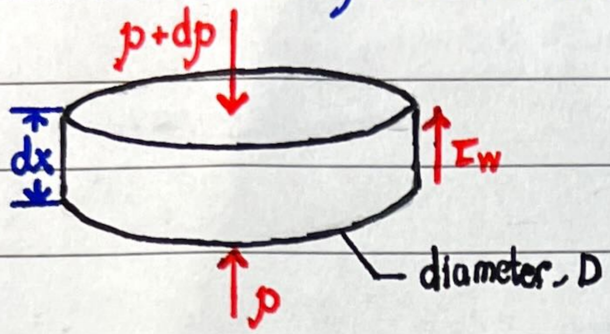


$$\frac{\pi D^2}{4}\frac{dp}{dx}dx = \pi D \tau_w dx$$

$$\downarrow$$

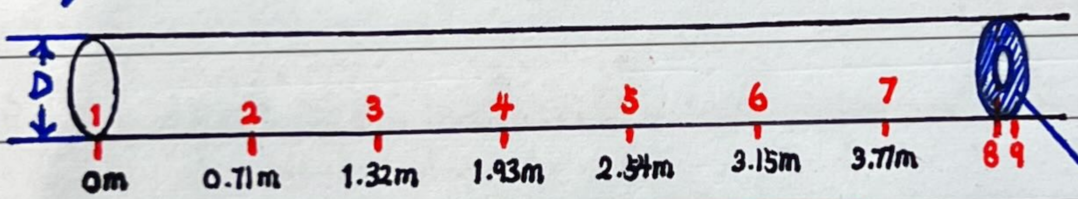
$$\tau_w = \frac{D}{4}\frac{dp}{dx}$$

$$Re_D = \frac{\rho V D}{\mu}$$

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho V^2} = f(Re_D)$$

$$\rightarrow \quad C_f = \frac{D}{4(\frac{1}{2}\rho V^2)}\frac{dp}{dx} = \frac{\Delta p}{\frac{1}{2}\rho V^2}\frac{D}{4L} = f(Re_D)$$

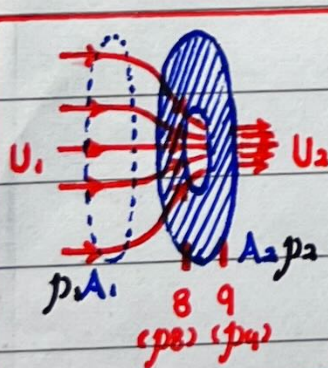
Experimental Facility and Procedure



0 pressure tapping

$D = 9.5mm$

orifuce plate



$A_2 < A_1 \xrightarrow{Q=\rho u A} U_2 > U_1$

$p_1 + \frac{1}{2}\rho U_1^2 = p_2 + \frac{1}{2}\rho U_2^2$

$\hookrightarrow p_1 < p_2$

$\hookrightarrow p_9 > p_8$ 突破梗阻

$\rightarrow k_{lam} = 0.10$

$\rightsquigarrow k_{turb} = 0.08$

$$\frac{1}{2}\rho_{air} V^2 = k\Delta p_0 = k(p_9 - p_8)$$

determine the mean velocity in the flow

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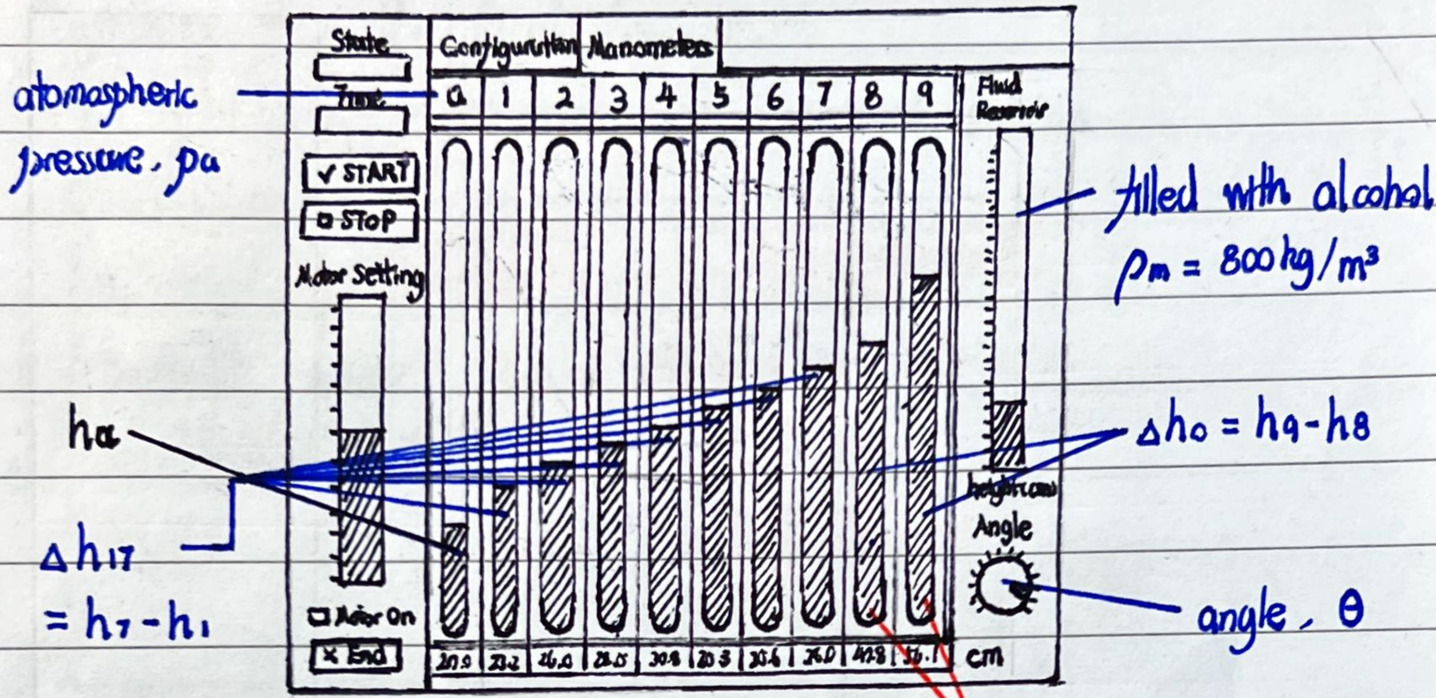
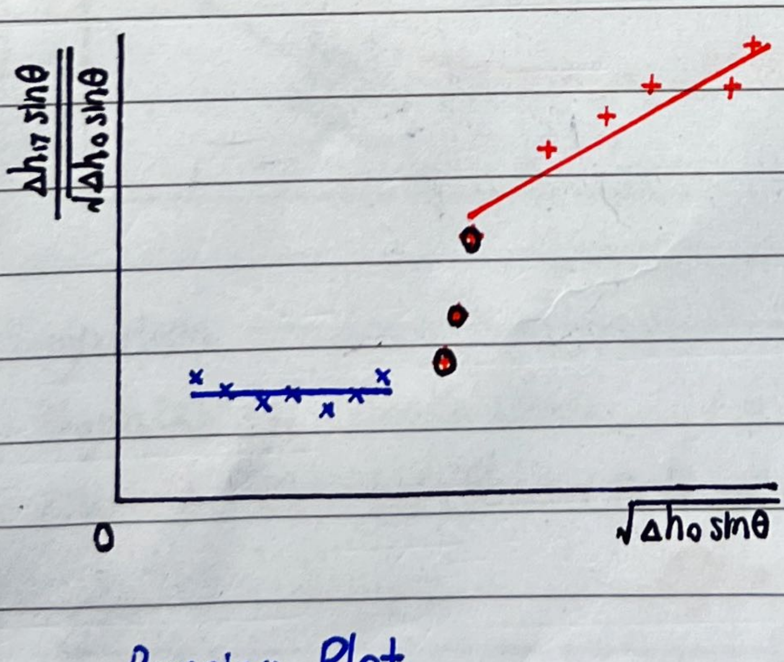


$$\boxed{\frac{1}{2}\rho_a V^2 = k\Delta p_0}$$

$$\boxed{\Delta p_0 = \rho g \Delta h = \rho_m g \Delta h_0 \sin\theta}$$



— Laminar Test Cases

o Transition Test Cases

+ Turbulent Test Cases

Running Plot

- isolate transition
- determine the distribution of points

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Preparing for the Oral Assessment

24h after lab → 30 min oral assessment session

Calculation

velocity $V = \sqrt{\frac{2k\Delta p_0}{\rho_a}} = \sqrt{\frac{2k\rho_m g \Delta h_0 \sin\theta}{\rho_a}}$

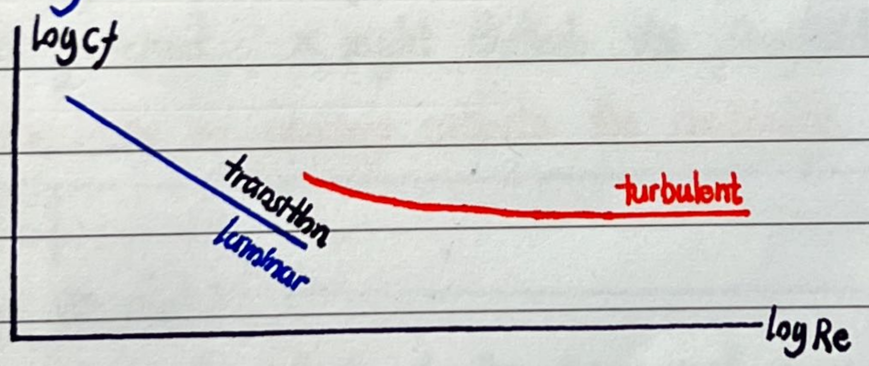
Reynolds number $Re_D = \frac{\rho V D}{\mu} = \frac{\rho_a V D}{\mu_{air}}$

skin-friction coefficient $C_f = \frac{\Delta p}{\frac{1}{2}\rho_a V^2} \frac{D}{4L}$

$= \frac{\rho_m g \Delta h \sin\theta}{k \rho_m g \Delta h_0 \sin\theta} \frac{D}{4L}$

$= \frac{\Delta h}{k \Delta h_0} \frac{D}{4L} = \frac{1}{4k} \boxed{\frac{d(\frac{h}{h_0})}{d(\frac{x}{D})}}$ gradient

Plotting



Determine the slope of the straight line

Comparison

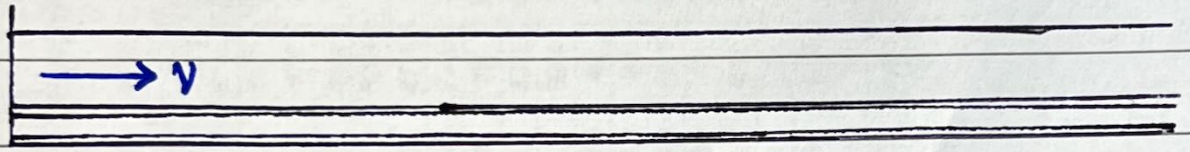
Hagon [2] and Poiseuille (1840) $C_f = \frac{16}{Re_D}$

Nikuradse (1932) and Stanton & Pannell (1914)

$$\frac{1}{\sqrt{C_f}} = -0.4 + 4\log(Re_D\sqrt{C_f})$$

11/21/2022 Pipe-flow During Lab

- $P_{atmosphere}$ = 989.9 mbar = 98990 Pa
- T = 24.2°C = 297.35 K



* pressure tupper 4 do not exist in equipment

ignore the value record by tapping 4

* value is unstable at first few seconds in steady flow

* value cannot keep in a constant value in unsteady flow

just approximate it at average

Other values

μ_{air} at 24.2°C ≈ 1.84×10^{-5} kg/ms²

$\rho_a = \frac{P_a M}{RT} = 1.15967492$ kg/m³

$\rho_m = 800$ kg/m³

$D = 9.5$ mm $= 9.5 \times 10^{-3}$ m

fluid reservoir height

h = 20 cm

* μ_{air} is estimated value from diagram standard

11/21/2022 Pipe-flow During Lab

Number of attempts	Motor Setting (V)	Angle (degrees)	Fluid Reservoir height (cm)	Pressure Tapping	a	1	2	3	4*	5	6	7	8	9
				Position x (m)										
1				h (cm)										
2														
3														
4														
5														
6														
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19												58.9	63.4	59.2

*Ignore pressure tapping cause it is not exis

* Data need to be processed again with the error occurs at ground state

11/21/2022 Pipe-flow During Lab

Number of attempts	Motor Setting (V)	Angle (degrees)	Fluid Reservoir height (cm)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			

Pressure Tapping	a	1	2	3	4	5	6	7	8	9
x (m)										
error (m)										
h (m)										

* Some value might be invalid due to the inaccurate estimate from reading

11/21/2022 Pipe-flow Post-Lab

Number of attempts	Δh_0 (m)	Δh_{17} (m)	$(\Delta h_0 \sin\theta)^{1/2}$ ($m^{1/2}$)	$\Delta h_{17}\sin\theta$ (m)	$\Delta h_{17}\sin\theta / (\Delta h_0 \sin\theta)^{1/2}$ ($m^{1/2}$)	Pressure Tapping						
						x_1-x_1 /1000D	x_2-x_1 /1000D	x_3-x_1 /1000D	x_4-x_1 /1000D	x_5-x_1 /1000D	x_6-x_1 /1000D	x_7-x_1 /1000D
						h_1-h_1 /Δh_0	h_2-h_1 /Δh_0	h_3-h_1 /Δh_0	h_4-h_1 /Δh_0	h_5-h_1 /Δh_0	h_6-h_1 /Δh_0	h_7-h_1 /Δh_0
2												
3												
4												
5												
6												
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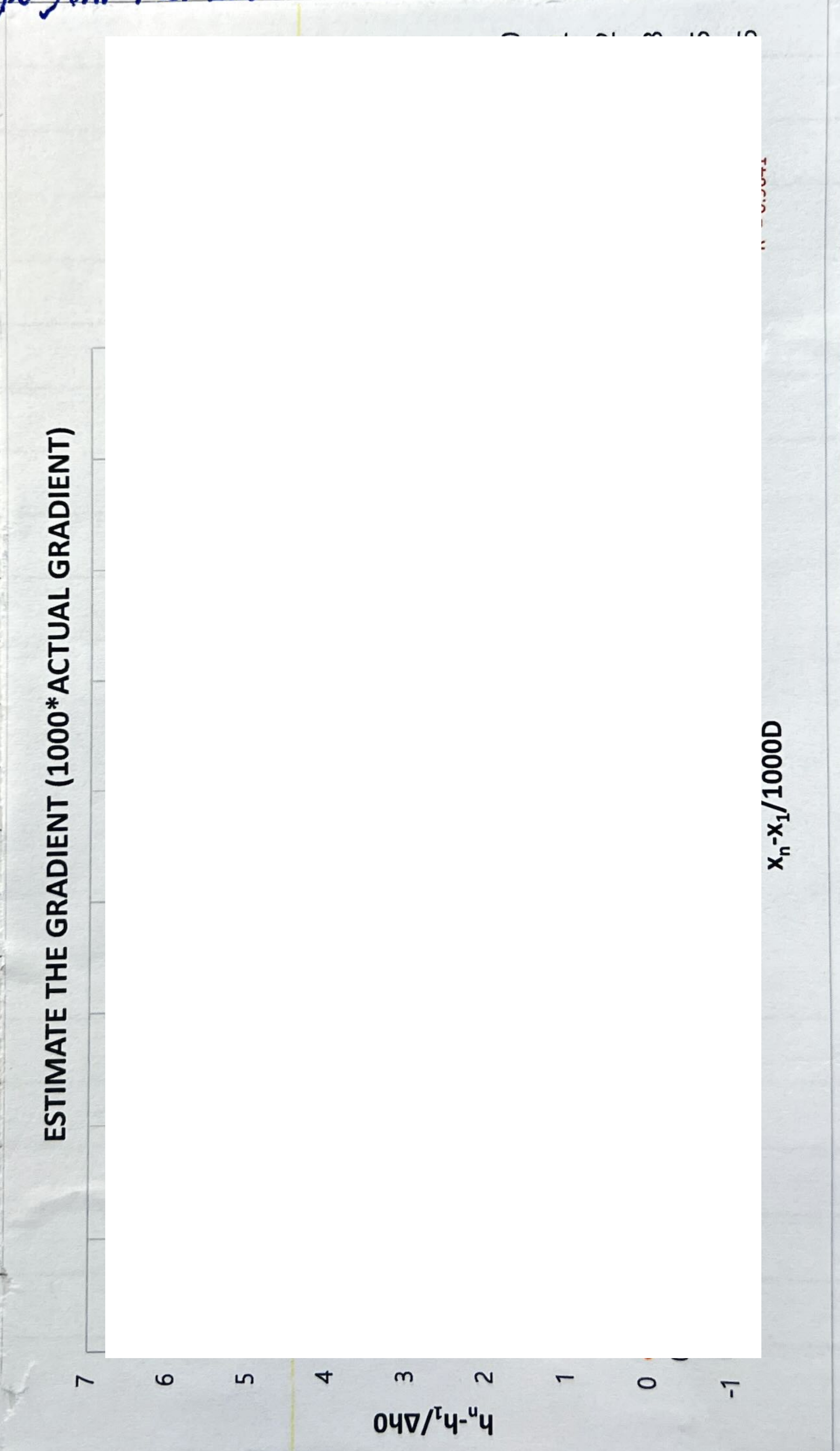
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												

11/21/2022 Pipe-flow Post-Lab

Find gradient
to estimate
the Cf

Cf =
(1/4) gradient

Laminar and
turbulent flow
have much
higher gradient
than transition
flow, and
laminar flow
have very
large-disturbed
data point



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Number of attempts	Motor Setting (V)	Type of Flow	V (m/s)	Re_D	τ_w (Pa)	C_f	C_f comparison			
							Hagon and Poiseuille		Nikuradse and Stanton&Pannell	
			$(2k\Delta p_0/\rho_a)^{1/2}$	$\rho_a VD/\mu$	(D/4)(dp/dx)	(1/4k)gradient	$16/Re_D$	Δ	$1/(Cf)^{1/2}=-0.4+4\log(ReD(Cf)^{1/2})$	Δ
1	0									
2	9.3									
3	9									
4	8									
5	7									
6	6									
7	5									
8	4									
9	3									
10	2									
11	1									
12	5									
13	5.46									
14	0.5									
15	1.25									
16	1.5									
17	0.35									
18	0.62									
19	0.72									

* Attempts 14, 16, 18, 19 is invalid caused by the measurement error

Type of flow can be determined through the running plot in next page →

k lam = 0.10

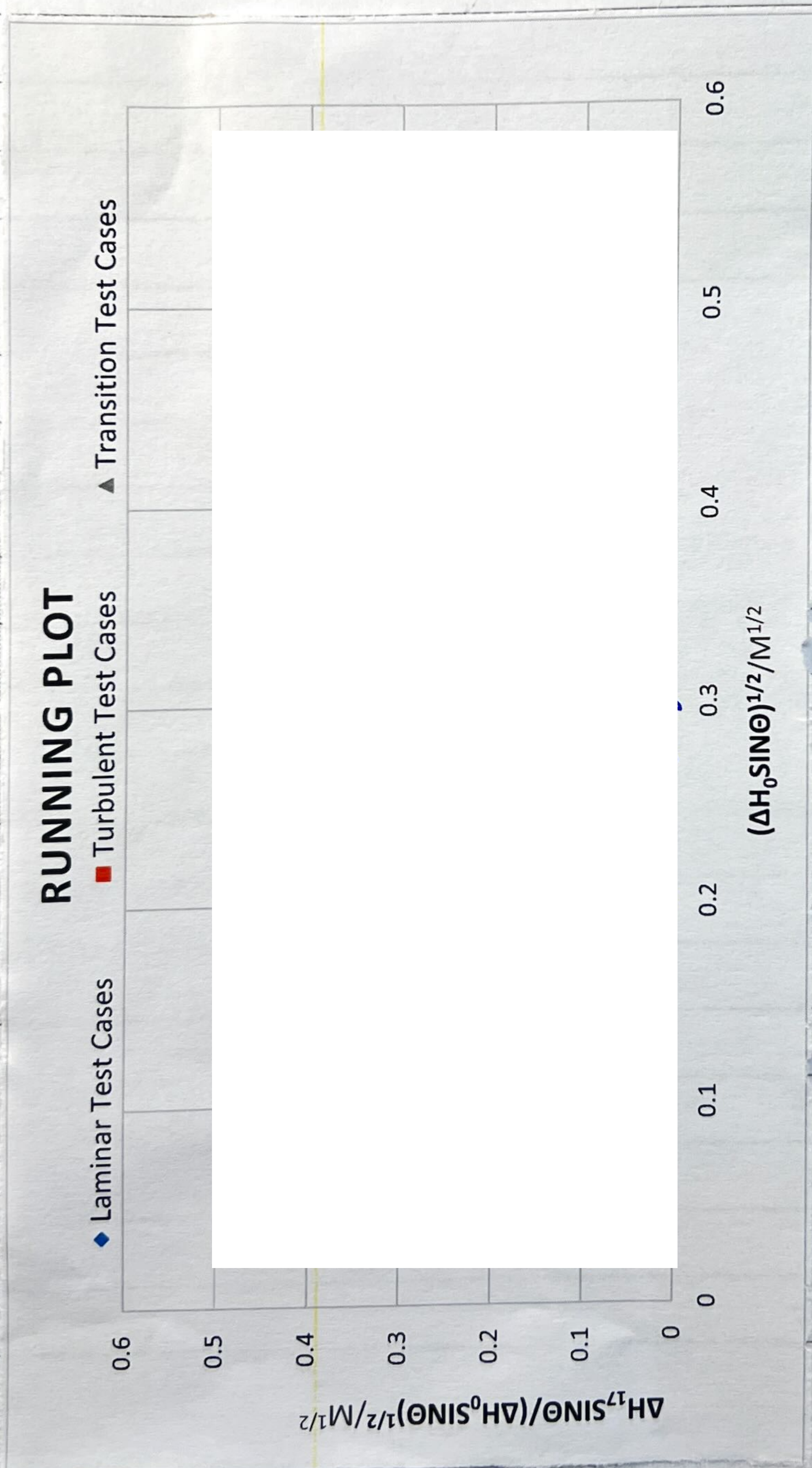
k turb = 0.08

Assume that k tran = 0.08

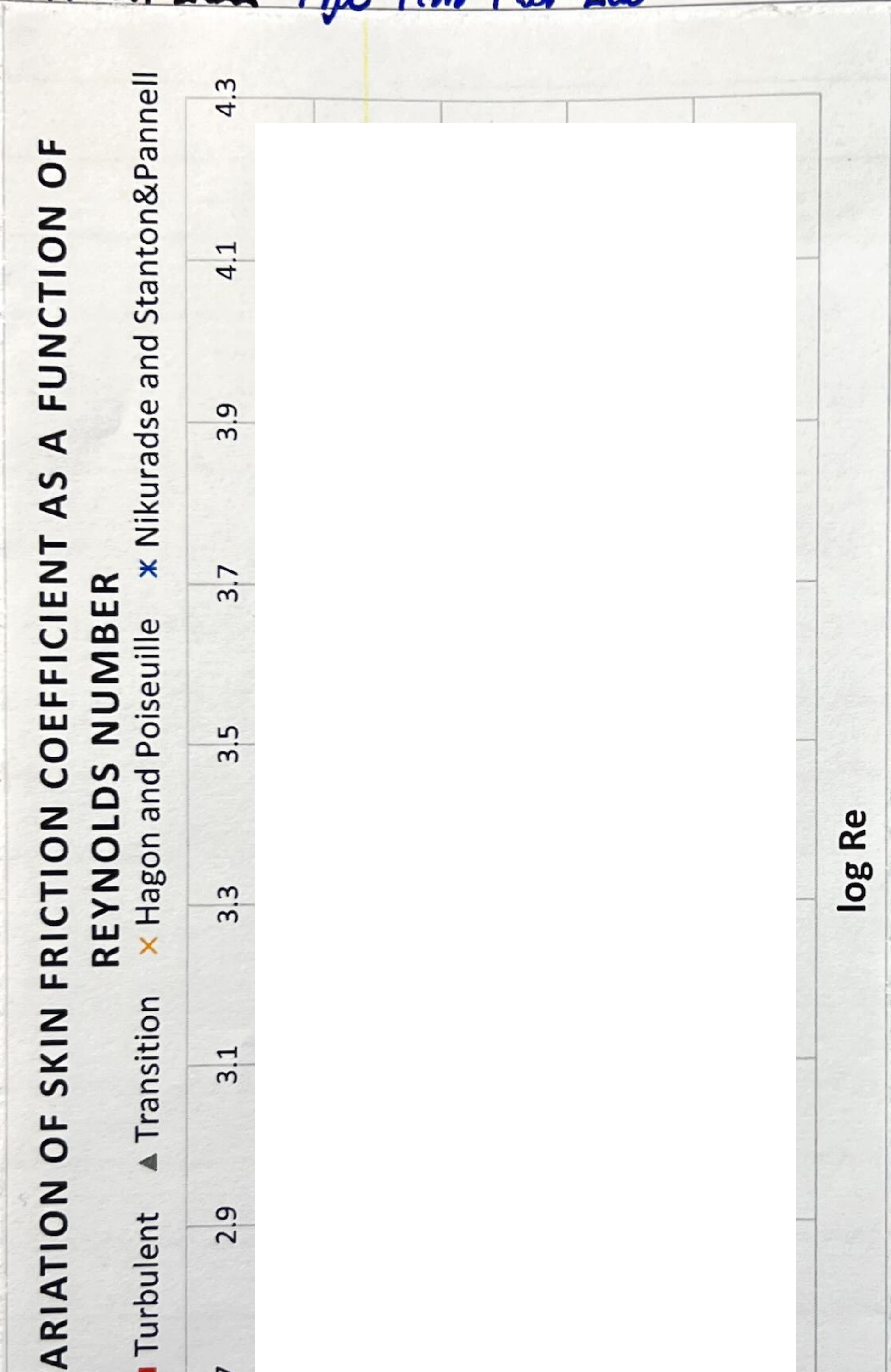
Δ might caused by the energy dissipation

Δ might caused by the measurement error

11/21/2022 Pipe-flow Post-Lab



11/22/2022 Pipe Flow Post-Lab



× Hagon and Poiseuille use to calculate the laminar flow slightly higher than experimental value

× Nikuradse and Stanton & Pannell use to calculate the turbulent flow slightly higher than experimental value

The gap is shown by the Δ in the graph in previous page